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Sunscreen Science: How Thick Should You Apply?

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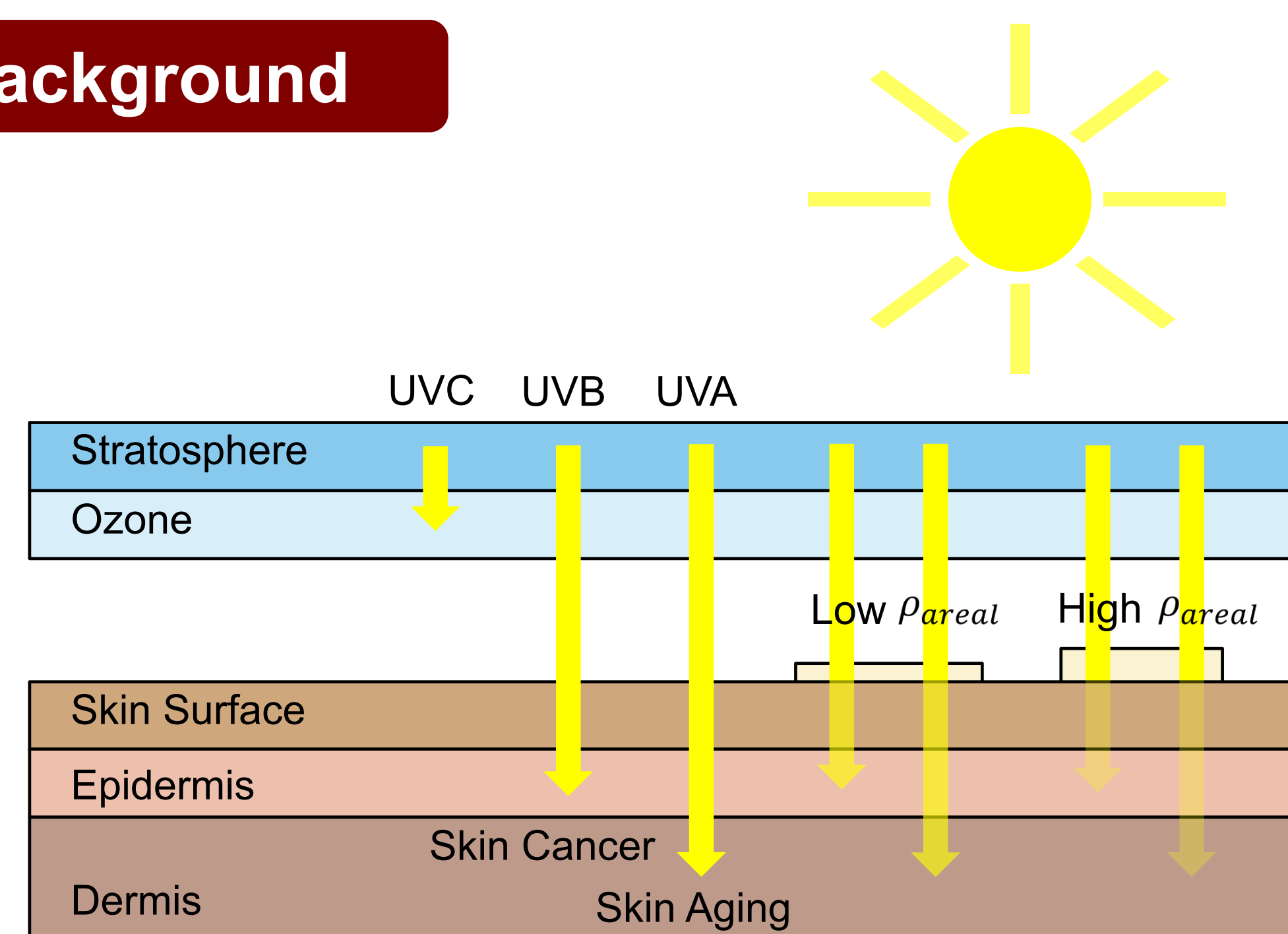
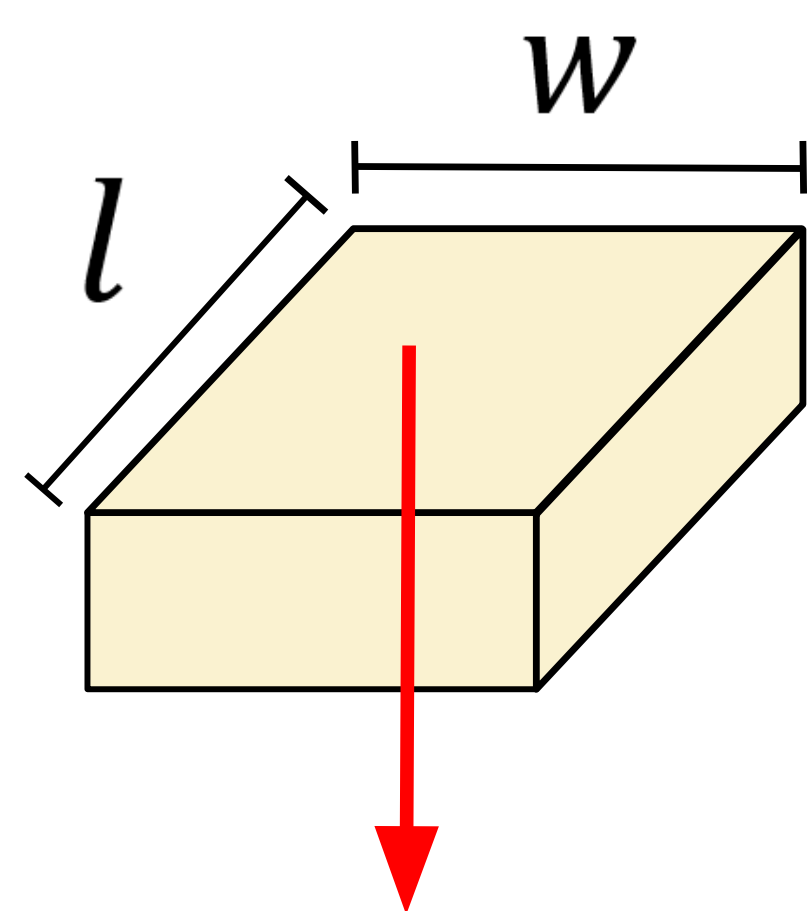
2.671 Measurement and Instrumentation

Abstract

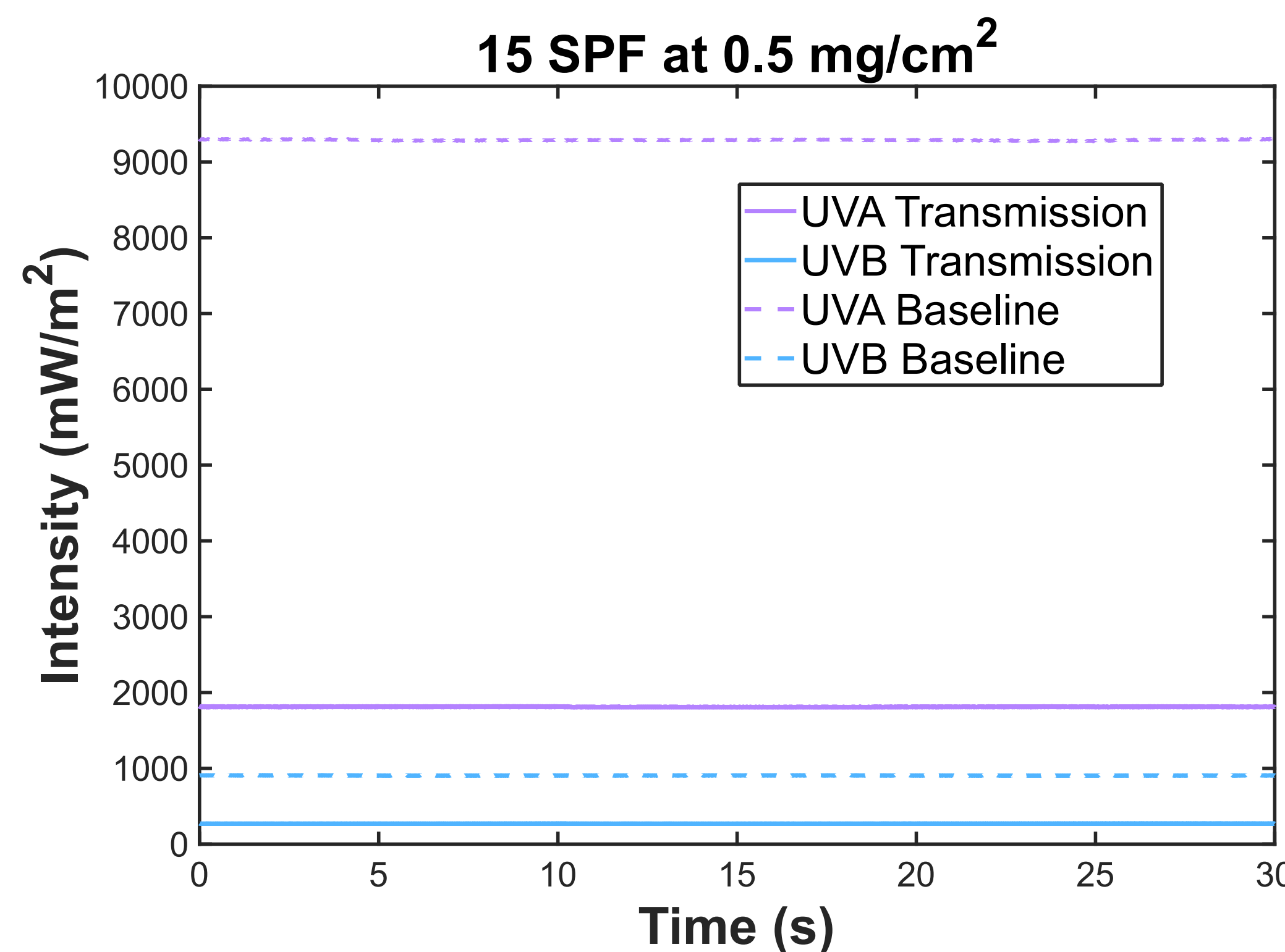
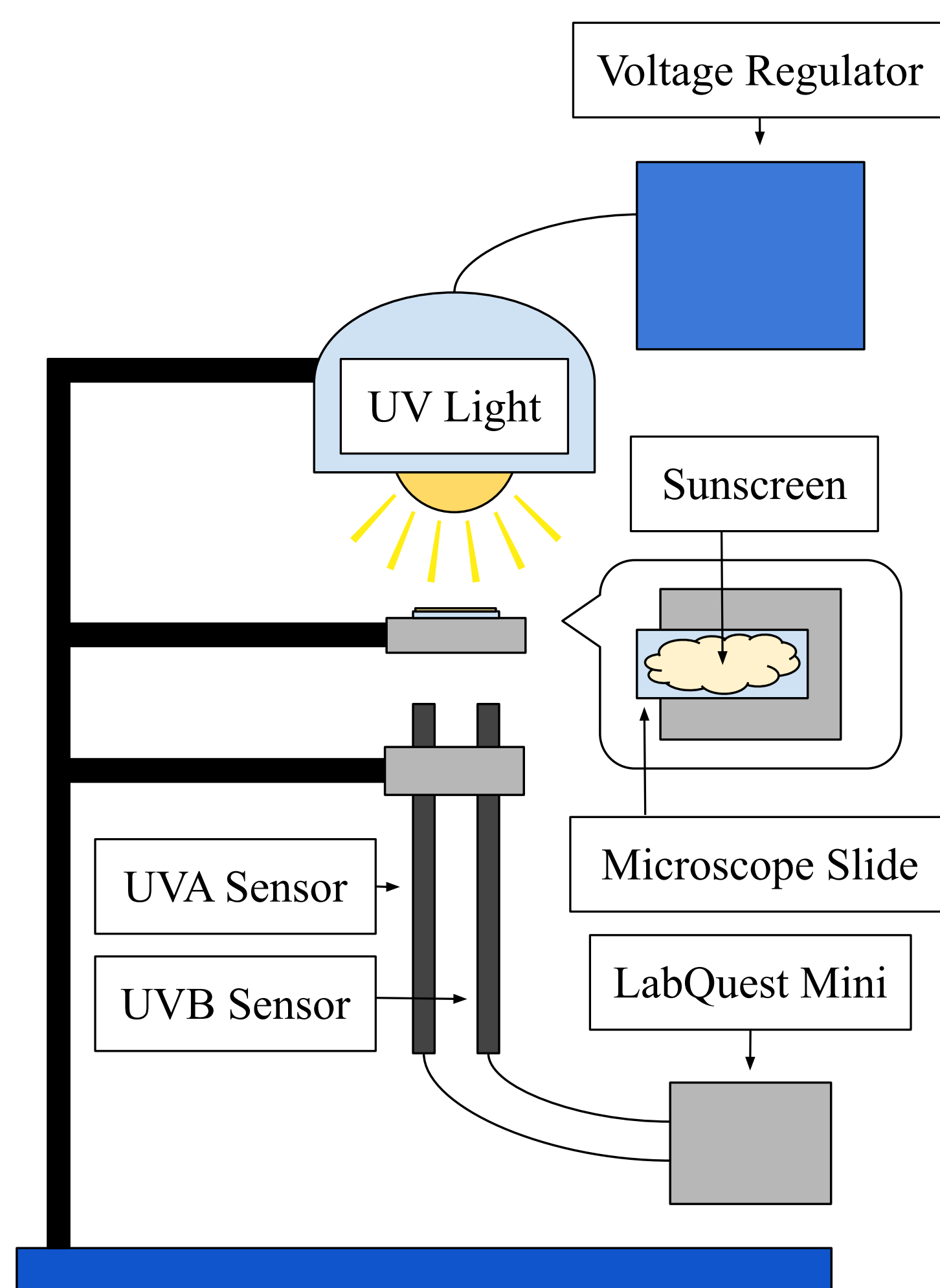
The interaction between the Sun Protection Factor and areal density of sunscreen, or the weight per unit area, is a critical but not well understood factor in decreasing UV transmission. To understand how SPF impacts the relationship between areal density and both UVA and UVB transmission, the influence of SPF on the rate of UV attenuation with increasing areal density was characterized. UV transmission through four different SPFs of a widely used sunscreen applied at varying areal densities was measured with UV sensors. UV transmission decreased exponentially with areal density, and the attenuation coefficients, determining the rate of attenuation, increased logarithmically from the lowest to highest SPF by $132 \pm 38\%$ for UVA and $75 \pm 23\%$ for UVB. Consequently, the higher SPF sunscreens approached maximum UV attenuation more quickly than the lower SPF sunscreens, suggesting that higher SPF may mitigate the harms of underapplication.

Background

$$\rho_{areal} = \frac{\text{Sunscreen Weight}}{l \times w}$$



Experimental Design



$$UV_{trans}(\%) = 100 \times \frac{UV_{transmission}}{UV_{baseline}}$$

Acknowledgements

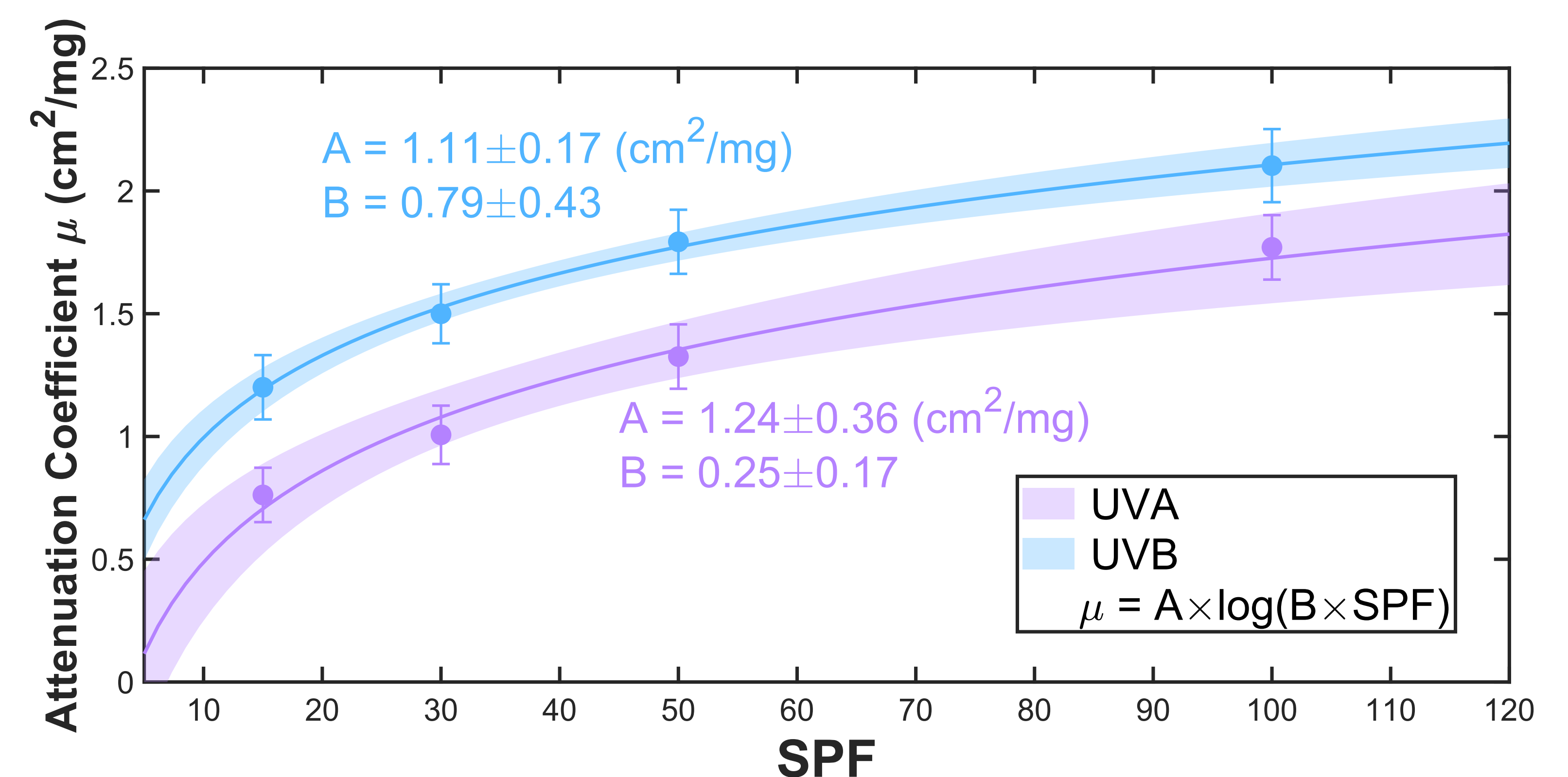
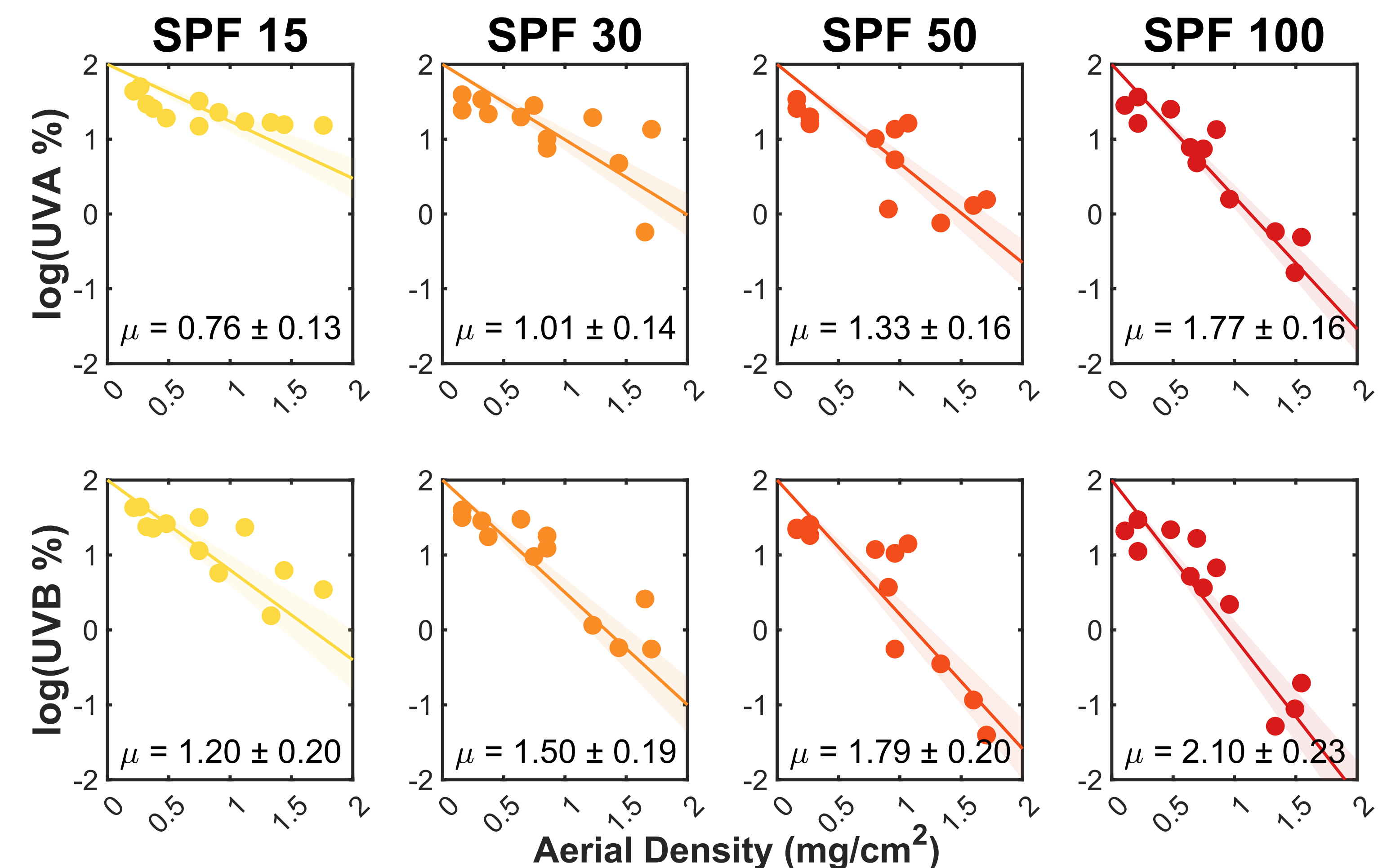
Thank you to Dr. Huang, Dr. Hughey, and Dr. Bruner for their support and guidance and to Christian Biggins, Marc Vidal, and Dominic Hess for their unwavering support.

References

[1] Amazon, . [Online]. Available: <https://www.amazon.com/Banana-Boat-Sunscreen-Performance-Spectrum/dp/B0033ZL5YY?th=1>. [Accessed: 22-Oct-2025].

Attenuation Coefficients

Beer-Lambert Law: $UV_{trans} = 100 \times 10^{-\mu \rho_{areal}}$



Conclusions

- The attenuation coefficients, determining the rate of UV attenuation, increased the lowest to highest SPF by $132 \pm 38\%$ for UVA radiation and $75 \pm 23\%$ logarithmically from % for UVB radiation.
- Consequently, the higher SPF sunscreens approached maximum UV attenuation more quickly with increasing areal density, suggesting higher SPF sunscreens may mitigate the harms of underapplication.
- The attenuation coefficients were larger for UVB than UVA radiation for all SPFs, with a largest difference of $58 \pm 29\%$, indicating that underapplication increases UVA radiation more quickly than UVB radiation.